Are MLB Players Challenging Optimally?

By Kendall Ruth

Summary:
With two strikes, a batter's challenge is overturned only 39.9% of the time, compared with 58.7% when there are no strikes. A defense that challenges with three balls sees 45.5% overturned, compared with 63.9% when there are no balls.

2026 regular season Automated Ball-Strike (ABS) challenge system data shows that batters and defenses are least selective in exactly these counts. Challenges with two strikes were overturned about 32% less often than challenges with zero-strikes. If two-strike challenges succeeded at the zero-strike rate, it could be estimated that an additional 106 runs would have been scored per season (approximately 3.5 per team per season). In a close playoff race, these additional runs could make a big difference.

Defensive challenges show a similar pattern, with the focus on balls rather than strikes. Challenges with three balls were overturned about 28.7% less often than challenges with no balls. Like batters, defenses appear less selective about which pitches to challenge when the batter is close to a walk. The same approach estimates about 69 additional runs would have been scored over the course of a season (approximately 2.3 per team per season).

Background:
The 2026 season is the first season MLB used the ABS challenge system, which uses the Hawk-Eye tracking systems that are already installed in stadiums. This system allows batters, catchers, and pitchers to challenge pitches that they believe were called incorrectly by an umpire. When a player challenges a pitch call, the system maps the pitch's location within the strike zone for the individual batter and either upholds the call or overturns it. The ABS challenge system was successful this season, with all teams using it multiple times a game, and many incorrect calls being overturned. I was interested in determining whether the athletes were using the ABS as effectively as possible.

Methodology:
This project focused on ABS calls during the 2026 season. Pitch by pitch data was obtained using MLB Statcast API. Additional variables, including pre pitch count, base state, and the number of challenges remaining were calculated from the available data. Pitch location and zone information were also retrieved using the PyBaseball library. The resulting dataframes were merged using game, at-bat, and pitch numbers to associate pitch locations with individual challenges.

Run expectancy, which was used in this project, was also calculated using the pybaseball library. Each play was separated by count and base state creating a matrix showing the average number

of runs scored at that point in an inning, showing how influential a challenge was in leading to more runs being scored.

Findings:
A logistic regression model was run for both Offensive (Batters) and Defensive (Pitchers + Catchers) challenges, to help identify the variables with the largest impact on the overturn rate. Discussed below are the most important variables in both offensive and defensive challenges.

Offensive:
Out of all the potential factors that would influence a pitch being overturned, the most important was the number of strikes in the count before the pitch (P-Val: 0.000). The second most significant is the current base state. The more strikes that there are in the count, the more likely it is for batters to challenge strikes that are further into the strike zone. Figure 1 below shows the data that validates this claim.

Figure 1: Challenge volume, overturn rate, and average miss distance by strike count

Num Strikes	Num Challenges	Num Overturned	Overturn Rate	Average Miss Dist (in)
0	1617	949	59%	1.91
1	1530	733	48%	1.56
2	1619	646	40%	1.28

After evaluating the relationship between strike counts and average miss distance, I decided to investigate this relationship further. Based on the individual miss distances for batter challenges, this expectation holds (Figure 2). Higher strike counts are associated with an increased number of challenges involving pitches near the boundary of the strike zone. To further refine what is happening to batters, the miss distance shrinks when batters reach two strikes.

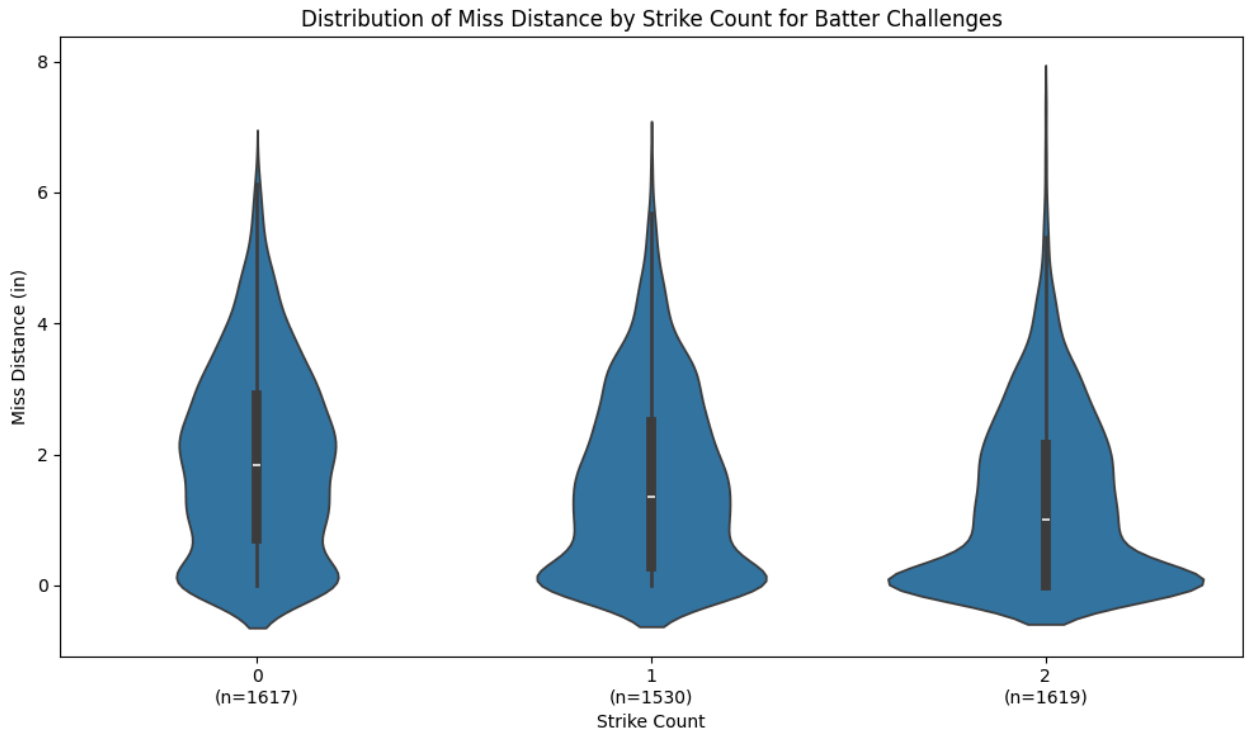


Figure 2: Violin plot of distribution separated by number of strikes

A specific example of a batter who challenged less effectively with two strikes was Zach Neto of the Angels. Neto had 22 challenges with the bases empty, split almost evenly across the three strike counts: 8 with no strikes, 7 with one strike, and 7 with two strikes. Similar to the overall league, even he was not immune to the curse of the challenge, especially with two strikes. With two strikes, he was overturned on just 1 of 7 challenges, compared with 6 of 8 with no strikes. With samples this small, one overturned challenge can move the rates by more than 10 points, so this example should be used as an example of the league-wide pattern, and not a measure of how well Neto makes ABS decisions. The average miss distance illustrates the same point, with the caveat that the average miss distance increased with 1 strike. Average miss distance is significantly lower with 2 strikes than with 0 strikes (0.94 in vs 2.07 in) (Figure 3).

Figure 3: Zach Neto challenge stats with the bases empty

Num Strikes	Num Challenges	Num Overturned	Overturn Rate	Average Miss Dist (in)
0	8	6	75%	2.07
1	7	5	71%	2.88
2	7	1	14%	0.94

Defensive:
Out of all the potential factors that would influence a pitch being overturned, the most important was the number of balls in the count before the pitch (P-Val: 0.000). The second most significant is the number of outs at the time of the challenge. The data shows that the more balls in the count, the more likely it is for catchers or pitchers to challenge balls that are further outside the strike zone. Figure 4 below shows the data that validates this claim.

Figure 4: Challenge volume, overturn rate, and average miss distance by strike count

Num Balls	Num Challenges	Num Overturned	Overturn Rate	Average Miss Dist (in)
0	2424	1549	64%	1.18
1	1514	877	58%	1.33
2	1056	568	54%	1.49
3	797	363	46%	1.72

There are two main differences between the defensive challenges table and the batter challenges table. The first is that the number of challenges is not as evenly split as it was for the batter challenges. For batters, there was a virtually even split between each possible number of strikes. For catchers and pitchers, that is not the case; there are more challenges with zero balls than with any other number of balls. The second difference is that the average miss distance is further from the strike zone. For the defense, this is to be expected, as it means they are challenging more obvious balls with three strikes than with zero strikes.

By looking at the distribution of the pitches challenged by the defense in Figure 5, all these points are proven. Pitchers and catchers become less selective and are more willing to challenge any pitch they deem is close when there are more balls in the count.

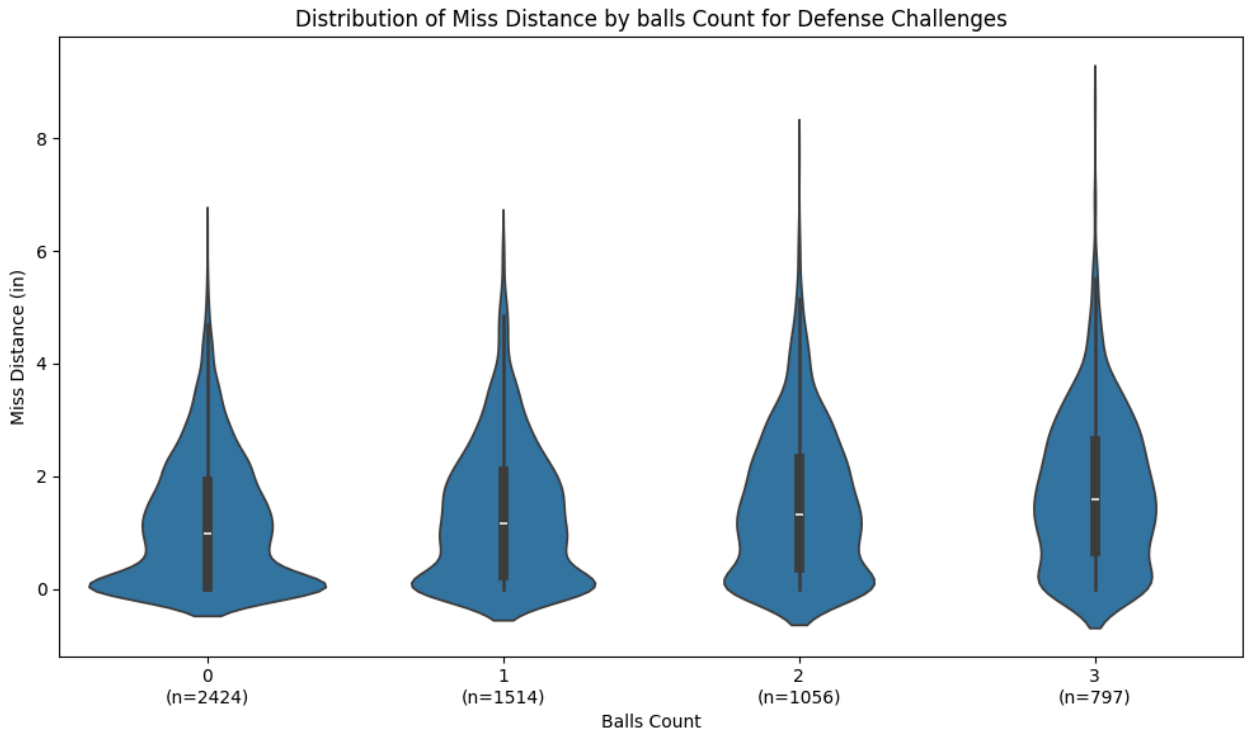


Figure 5: Violin plot of distribution separated by number of balls

One example of this overall trend of challenges is Austin Wells, a catcher for the New York Yankees. Wells' example follows a similar pattern to Neto's that was shared earlier. While the number of challenges he had with the bases empty was not quite as even, the overturn rate decreased as the number of balls increased (82% vs 67% vs 62% vs 36%). The average miss distance is larger with 3 balls than with zero balls, and it generally decreases (1.11 in vs 0.99 in vs 1.32 in vs 2.21 in), but with a count of one ball, the average miss distance is slightly lower. (Figure 6) Overall, Wells' statistics are a good example illustrating that catchers are also less selective with challenges as the number of balls in the count increases.

Figure 6: Austin Wells' challenge stats with the bases empty

Num Balls	Num Challenges	Num Overturned	Overturn Rate	Average Miss Dist (in)
0	39	32	82%	1.11
1	12	8	67%	0.99
2	16	10	63%	1.32
3	11	4	36%	2.21

Limitations:

There are a few limitations to this study. The biggest limitation is that it requires buy-in from both the player and the coach. For most players/coaches, it makes the most sense to challenge

any close pitch with two strikes or three balls. Getting them to buy into the idea that challenging with zero strikes and balls is better might be difficult for some athletes to believe, even after data is shown to them. Another limitation is that this was only one season of data, and it is possible that with more seasons of the ABS challenge system in use, these findings would change. A final limitation is that the value of 3.6 runs per team is average. If investigated further, these run numbers may be higher on an individual team basis.

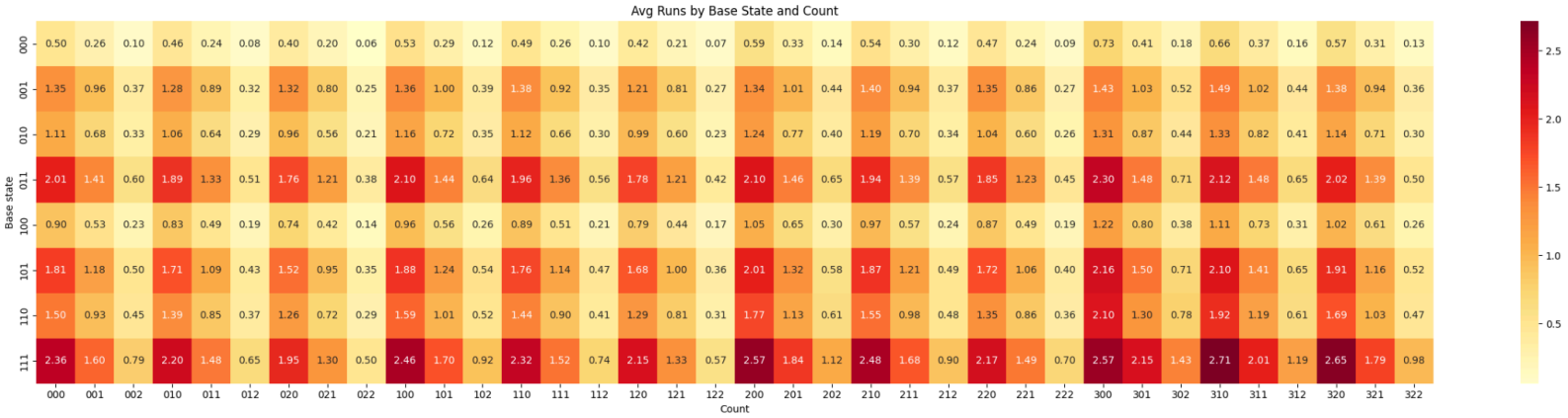
Data Notes:
The number of 2026 challenges data that was queried from the MLB Statcast API matches the totals on Baseball Savant's ABS Page with 10,557 challenged pitches. One challenge is classified differently: the MLB Stats API records that challenge as a confirmed batter challenge, whereas savant records it as an overturned fielder challenge. The cause of the difference is not confirmed. This accounts for our overturned total being one lower (5,685 vs. 5,686). This report uses the API classification.

Implications:
These findings would be of interest to almost anyone involved with a MLB team, including coaches and front-office staff. This analysis suggests that if hitters spend more of their challenges with zero strikes and catchers/pitchers with zero balls, then teams would score more runs. Making these small changes to a team's ABS Challenge approaches should be fairly easy, easier than many other changes a team could make. Along with this, they don't require any additional investment in the roster. Challenges can be quantified and evaluated to assess their effectiveness.To further this project, I plan to take the ideas from the basic evaluation and adapt them into a pipeline to generate a frequent internal report. This internal report would be sent to coaches and other interested stakeholders, allowing them to see, on both an individual and team basis, how well challenges are progressing.

Conclusion:
Across the 2026 MLB regular season, ABS challenges were overturned less often in the counts where a batter is close to a strikeout or to a walk. For offensive challenges, this is with two strikes and for defensive challenges this is with three balls. While the estimated run value of making these changes is modest on a team basis, these patterns add up across the whole league and could make a difference in a tight race. Future work would be to build this analysis into a pipeline that updates daily, with a framework for scoring each challenge by its run value gained or lost, so that challenges can be evaluated directly.

Appendix A: Run Expectancy Chart

Here is the run expectancy chart that I calculated. This is using data from 2021 – 2025 and calculating the number of runs that were scored in that half inning from that point on.



Appendix B: Full Logistic Regression Model Output For Batters

Below is the logistic regression model output showing the full output for the importance of each of the variables for batter challenge success rate.

Variable	Coefficient	Std Error	Z	P>|z|	Odds Ratio
Intercept	0.201	0.1267	1.5861	0.1127	1.2226
C(bucketed_base_state)[T.RISP]	-0.2349	0.0714	-3.288	0.001	0.7906
C(bucketed_base_state)[T.Runner on First]	-0.2404	0.0795	-3.0224	0.0025	0.7863
prev_balls	0.0065	0.0303	0.2145	0.8302	1.0065
prev_strikes	-0.3944	0.0399	-9.8797	0.0	0.6741
prev_outs	0.0171	0.0372	0.4588	0.6464	1.0172
challenges_available	0.1357	0.064	2.1213	0.0339	1.1453

Appendix C: Full Logistic Regression Model Output For Defenses

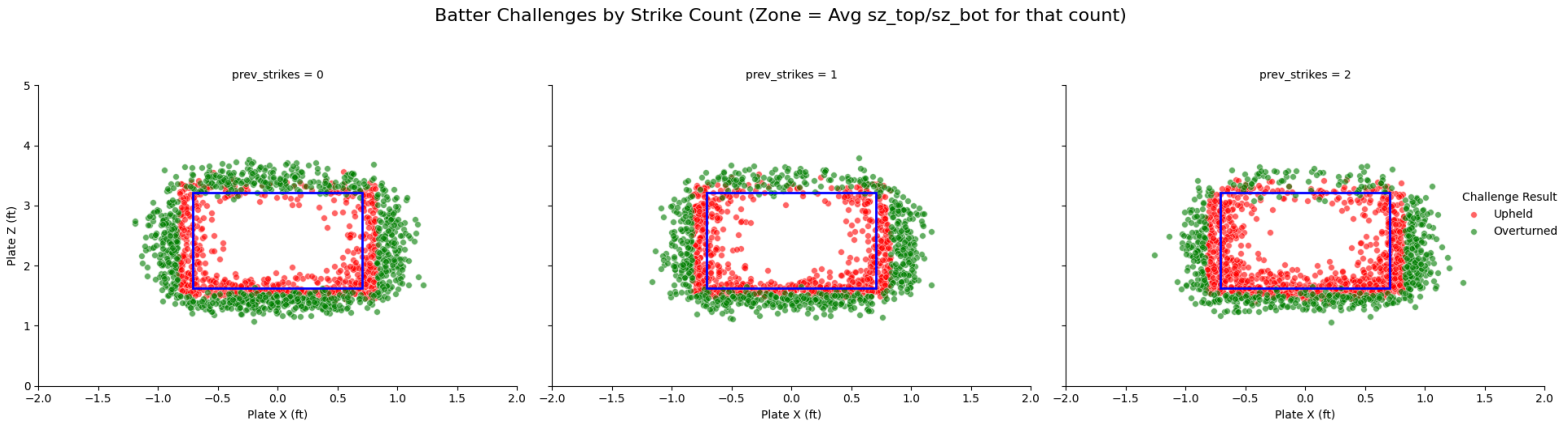
Below is the logistic regression model output showing the full output for the importance of each of the variables for defense challenge success rate.

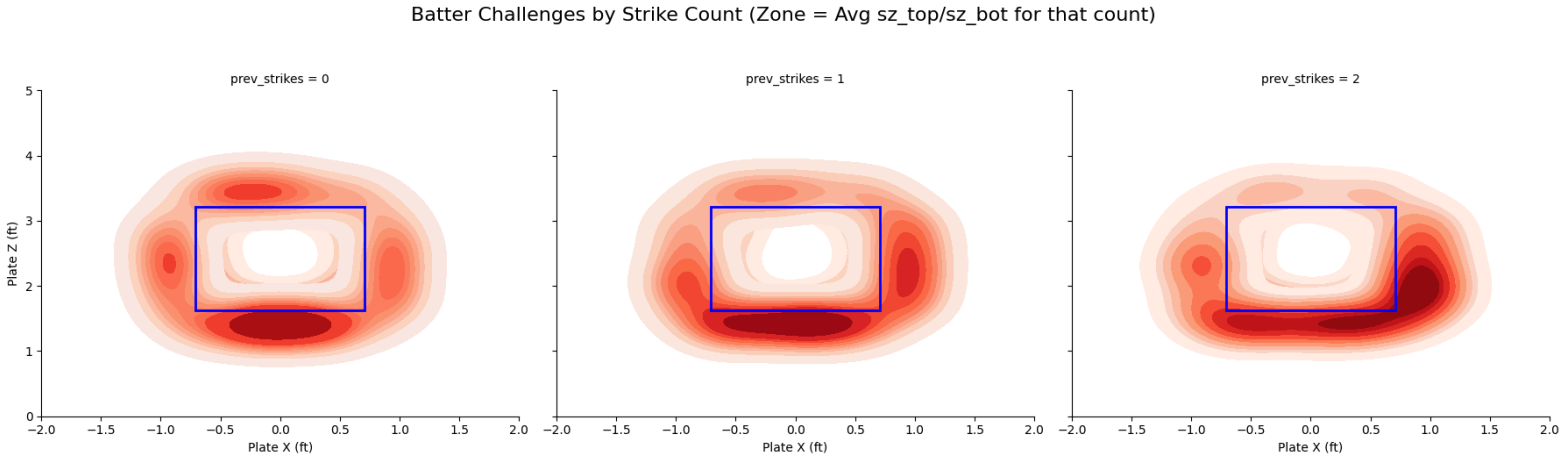
Variable	Coefficient	Std Error	Z	P>|z|	Odds Ratio
Intercept	0.8333	0.1251	6.6633	0.0	2.3008
C(bucketed_base_state)[T.RISP]	-0.1922	0.0647	-2.9695	0.003	0.8252
C(bucketed_base_state)[T.Runner on First]	-0.0576	0.0716	-0.8047	0.421	0.944

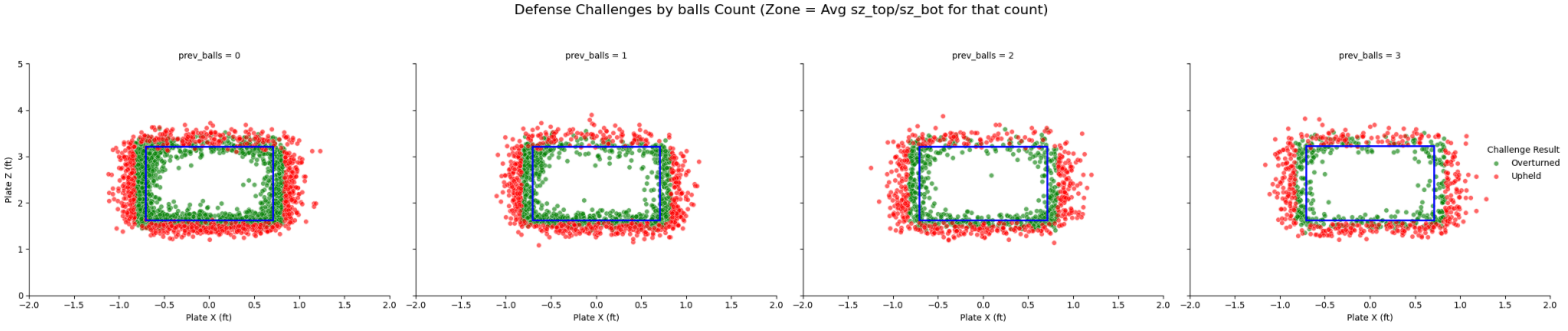
prev_balls	-0.2264	0.0278	-8.1283	0.0	0.7974
prev_strikes	-0.04	0.0358	-1.1178	0.2636	0.9608
prev_outs	-0.1532	0.0342	-4.4854	0.0	0.858
challenges_available	-0.0022	0.062	-0.0358	0.9715	0.9978

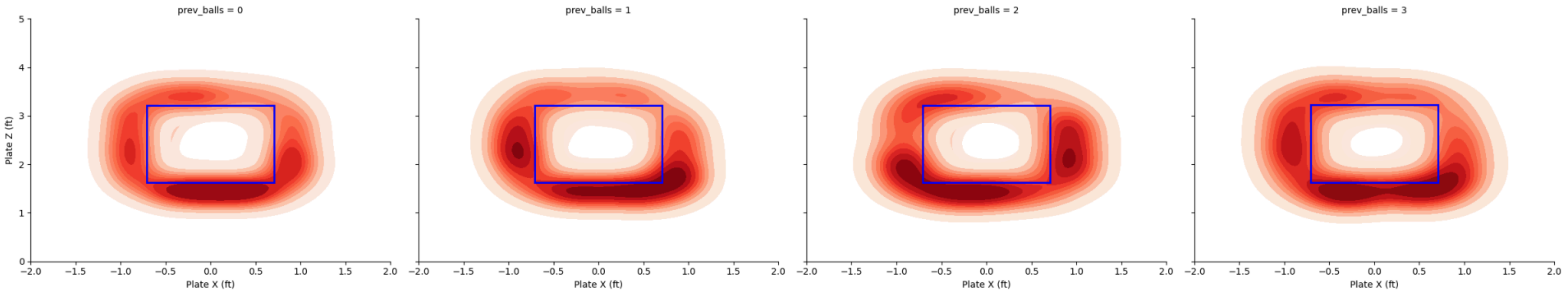
Appendix D: Other Visualizations

Below I have included other visualizations that I did not include in the main part of the report.








Defense Challenges by balls Count (Zone = Avg sz_top/sz_bot for that count)
prev_balls = 0
prev_balls = 1
prev_balls = 2
prev_balls = 3
Plate Z (ft)
Plate X (ft)
Plate X (ft)
Plate X (ft)
Plate X (ft)